MAJOR PROJECT

PROJECT SYNOPSIS

AI-Driven Multimodal Step-Up Authentication System with Real-Time Deepfake Detection

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Document	Project Synopsis

1. Background and Abstract

1.1 Background

The FinTech sector has shifted rapidly toward fully digital, remote-first customer interactions, with online banking, mobile wallets, and peer-to-peer transfer apps replacing in-branch verification. This convenience has expanded the attack surface available to fraud, as attackers now target weaknesses in remote identity verification rather than physical documents or in-person impersonation.

Generative AI has simultaneously advanced to the point where synthetic media can convincingly imitate a real person's face and voice. Deepfake video generation can superimpose a target's likeness onto a live or recorded feed, directly threatening camera-based Know Your Customer (KYC) and facial-verification checks, since a photo, a looped video, or a live face-swap can be presented in place of a genuine subject. Voice-cloning models can likewise synthesise a convincing replica of a person's voice from a few seconds of reference audio, enabling attackers to defeat voice-biometric checks or impersonate customers over the phone.

The SMS-based One-Time Password (OTP), the most common second factor in Indian digital banking, is similarly fragile: SIM-swap fraud, SS7 interception, and real-time phishing relays can all intercept an OTP intended for the legitimate account holder. Because OTP is typically a single, static checkpoint that does not adapt to transaction risk, a compromised OTP channel is often sufficient, on its own, to authorise a fraudulent high-value transfer. These three converging threats motivate an authentication approach that does not depend on any single static credential, but instead combines multiple modalities, evaluates risk contextually, and applies real-time liveness and authenticity checks that are difficult to spoof with pre-recorded or synthetic media.

1.2 Abstract

This project proposes an AI-Driven Multimodal Step-Up Authentication System that strengthens FinTech transaction security through contextual risk evaluation combined with real-time deepfake and voice-clone detection. Rather than applying uniform friction to every transaction, the system computes a composite risk score from signals such as transfer amount, geolocation deviation, and device velocity. Only transactions exceeding a defined risk threshold trigger a "Look & Speak" step-up challenge: the live camera feed is checked for natural liveness cues (blinks, head movement, reticle-guided alignment), while the live voice is matched via speech recognition against a fresh one-time spoken code and simultaneously screened for synthetic-voice artefacts. An LLM-based risk engine fuses the vision verdict, the audio match, the authenticity score, and the transaction risk into a single explainable decision of Allow, Escalate, or Deny, permanently recorded in an audit log. The prototype uses Next.js with WebRTC capture, a Django REST Framework backend, MediaPipe for facial landmarks, Whisper for speech recognition, PostgreSQL for storage, and Redis for session management, targeting sub-three-second verification with a fail-closed security posture.

2. Problem Statement and Core Objectives

2.1 Problem Statement

Contemporary FinTech platforms rely mainly on static, single-factor or loosely-coupled authentication designed before generative AI could produce convincing synthetic faces and voices. Facial verification often accepts a single still image or an unguided clip, defeated by a printed photo, a replayed video, or a face-swap deepfake. Voice verification based on static voiceprint matching is similarly vulnerable to cloned playback. SMS OTP offers no guarantee that the recipient device is genuinely held by the account owner and is routinely defeated by SIM-swap fraud. Nearly all deployed systems also apply identical friction regardless of transaction risk, causing either unnecessary inconvenience or insufficient scrutiny. There is a clear need for a layer that evaluates contextual risk first, combines independent modalities that must all be defeated simultaneously, and incorporates real-time liveness and synthetic-media detection.

2.2 Core Objectives

- Design a contextual, risk-based step-up mechanism invoked only when transaction risk crosses a defined threshold, minimising friction for routine activity.
- Build real-time visual liveness detection (blinks, head movement, reticle alignment) to defeat photo, video, and deepfake spoofing.
- Build an audio pipeline combining speech-recognition code matching with voice-authenticity scoring to defeat cloned or replayed voice.
- Fuse multimodal signals via an explainable LLM-based risk engine that outputs a decision with a plain-language rationale.
- Maintain a tamper-evident audit trail for compliance and dispute resolution.
- Keep end-to-end verification latency under three seconds for usability.
- Adopt a fail-closed posture: pipeline failures or ambiguous scores default to denial or escalation, never silent approval.
- Ensure the interface meets WCAG 2.1 accessibility guidelines, with a fallback path for users without camera/microphone access.

3. System Methodology and Flow

3.1 The "Look & Speak" Verification Process

The methodology centres on a two-stage challenge called "Look & Speak", invoked only when the Transaction Risk Engine's contextual assessment (transfer amount, geolocation deviation, device velocity, and payee risk) crosses a configured threshold. Below the threshold, the transaction proceeds without friction; above it, the session escalates into the challenge.

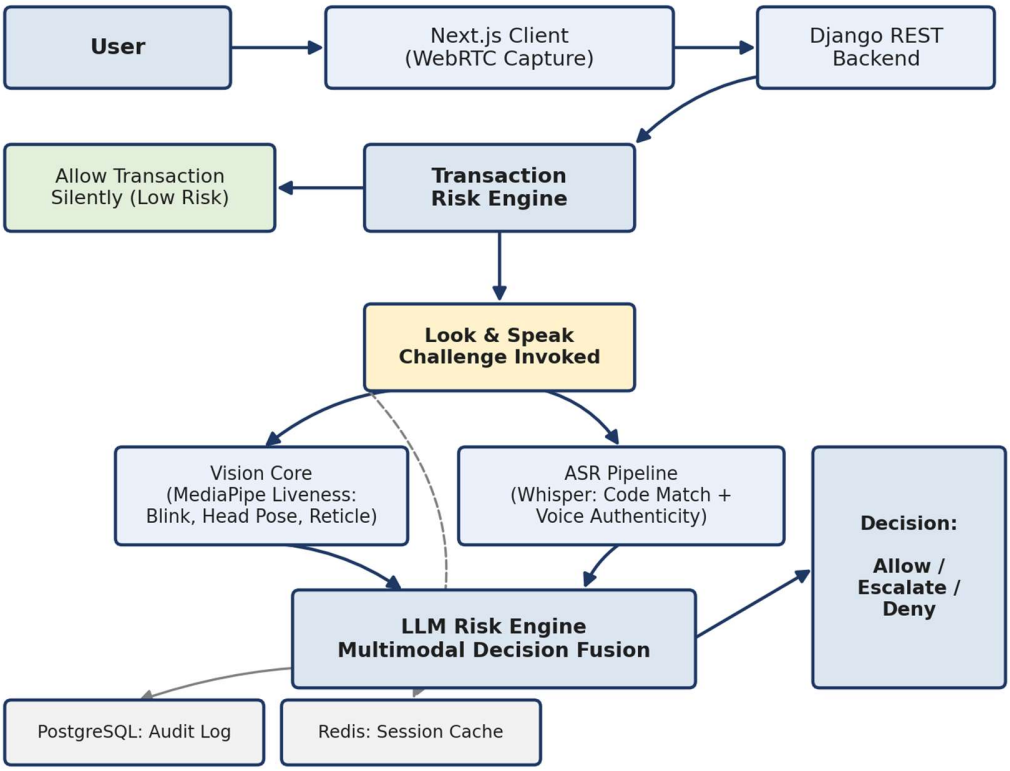
In the "Look" stage, the Next.js client opens a secure WebRTC session and displays a reticle guiding the user's face into frame. The Vision Core, built on MediaPipe, tracks the eye-aspect ratio for natural blinking and issues

a random head-movement prompt. Because these cues require a physically present, three-dimensional face responding to an unpredictable prompt, a static photo, replayed video, or real-time deepfake cannot easily satisfy them within the available window.

In the "Speak" stage, a short random one-time phrase is displayed, and the user reads it aloud. The Whisper-based ASR pipeline transcribes the audio and matches it against the phrase, defeating simple replay, while a parallel voice-authenticity classifier screens for spectral artefacts typical of synthetic speech. Both outcomes, together with the original risk score, pass to the LLM Risk Engine, which evaluates them step by step and produces Allow, Escalate, or Deny with a plain-language rationale, permanently logged for audit.

3.2 System Data Flow

The diagram below shows the data flow between the Next.js client, the Django REST backend, the Transaction Risk Engine, the Vision Core, the ASR Pipeline, the LLM Risk Engine, and the PostgreSQL/Redis data layer.



4. Technology Stack and Deliverables

4.1 Technology Stack Overview

Component	Role in the System
Next.js	Front-end framework for the transaction interface and Look & Speak screen, with native WebRTC support for camera/microphone capture.
Django REST Framework	Backend framework exposing REST endpoints and orchestrating the Vision Core, ASR Pipeline, and LLM Risk Engine.
WebRTC	Real-time protocol for capturing and streaming live audio/video from the browser with minimal latency.
MediaPipe	On-device ML framework used for real-time facial landmark extraction, enabling blink and head-pose liveness checks.
Whisper ASR	Open-source speech recognition model used to transcribe the spoken one-time challenge phrase.
PostgreSQL	ACID-compliant database persisting users, transactions, and the immutable audit log.
Redis	In-memory store for session state, challenge nonces, and rate limiting.

4.2 Project Deliverables

- Project Synopsis and SRS (this pair of documents), prepared per IEEE Std 830 conventions.
- A functional prototype: Next.js front end + Django REST backend, demonstrating end-to-end step-up verification.
- Integrated Vision Core (MediaPipe liveness) and ASR Pipeline (Whisper code match + voice authenticity).
- A decision fusion engine producing explainable Allow / Escalate / Deny outcomes.
- A PostgreSQL-backed audit log and a minimal administrator review dashboard.
- A consolidated test report and a final project report with viva-voce presentation.